

Simulation-Based Adaptive Cruise Control Using Deep Q-Learning in CARLA

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Abstract—This paper presents an Adaptive Cruise Control (ACC) system using Deep Q-Learning in the CARLA simulation environment. The proposed approach enables the system to learn optimal driving policies through interaction with dynamic traffic, prioritizing safety, comfort, and efficiency. Experimental results demonstrate robust performance in complex scenarios, outperforming rule-based systems. This study highlights the potential of reinforcement learning for enhancing autonomous vehicle control, with future work focusing on multi-agent interactions and real-world deployment.

Index Terms—Adaptive Cruise Control, Deep Q-Learning, CARLA Simulator, Reinforcement Learning, Autonomous Vehicles, Safety-Critical Scenarios, Autonomous Driving Metrics, ADAS

I. INTRODUCTION

Advanced Driver Assistance Systems (ADAS), particularly Adaptive Cruise Control (ACC), enhance road safety and driving convenience by automating speed adjustment to maintain safe inter-vehicle distances. Traditional ACC systems rely on predefined rules, limiting their adaptability in dynamic traffic scenarios such as sudden braking, lane changes, or adverse weather.

Reinforcement learning (RL) offers a promising solution to these challenges by enabling real-time decision-making through environmental interaction. Deep Q-Learning (DQL) extends this capability

by integrating neural networks with Q-Learning, allowing the agent to learn optimal policies without manual tuning. This approach is especially suited for ACC, where adaptability to complex, unpredictable conditions is critical.

In this work, we propose a DQL-based ACC system implemented in the CARLA simulator, a high-fidelity environment for autonomous driving research. Our agent dynamically adjusts speed by learning from interactions with traffic, prioritizing safety, efficiency, and passenger comfort. Key contributions include:

- A novel DQL framework for ACC that integrates real-time sensor data and addresses six safety-critical scenarios
- Comprehensive evaluation demonstrating superior performance over rule-based systems
- Open-source implementation bridging simulation and real-world deployment

II. LITERATURE REVIEW

The development of Adaptive Cruise Control (ACC) systems has evolved through three distinct generations, each addressing fundamental limitations in dynamic environment handling:

A. Traditional Rule-Based Approaches

Early ACC systems employed sensor fusion (radar/lidar) with proportional-integral-derivative (PID) controllers. Rajamani’s seminal work [1] demonstrated radar-based predictive control achieving 2.1s time headway at 100km/h, but struggled with nonlinear vehicle dynamics during abrupt maneuvers. Subsequent improvements by Li et al. [2] introduced traffic-density adaptive rules, reducing speed tracking errors by 33% in highway scenarios through Kalman filtering. However, these deterministic approaches showed limited generalizability, with collision rates exceeding 5% in complex urban environments [3].

B. Reinforcement Learning Revolution

The advent of deep reinforcement learning enabled data-driven policy optimization. Zhong et al. [4] pioneered DQN-based ACC achieving 1.8% collision rates in simulated highway scenarios, though their discrete action space (10% throttle increments) caused uncomfortable jerk ($>2.5\text{m/s}^3$). Recent advances by Wang et al. [5] combined twin delayed DDPG with prioritized experience replay, reducing emergency braking events by 40% through continuous action spaces. Nevertheless, these studies focused narrowly on car-following scenarios, neglecting critical edge cases like pedestrian crossings or adverse weather.

C. Simulation-Based Validation Paradigm

High-fidelity simulators like CARLA [9] enabled systematic testing of ACC robustness. The CARLA Autonomous Driving Challenge 2021 [6] revealed that 78% of RL-based ACC systems failed in rain conditions, highlighting simulation-reality gaps. Our work extends Kumar’s validation framework [7] by introducing six new safety-critical scenarios (Table ??), including low-friction roads ($\mu = 0.25$) and emergency vehicle yielding—scenarios absent in prior RL-ACC literature.

D. Research Gaps

Despite progress, three critical limitations persist in current literature:

- **Narrow Scenario Coverage:** 82% of ACC studies test ≤ 5 scenarios [8], missing safety-critical cases

TABLE I
COMPARATIVE ANALYSIS OF ACC APPROACHES

Method	Collision Rate	Comfort (Jerk)	Test Scenarios
Rule-Based [1]	5.2%	2.1 m/s ³	3
DQN [4]	1.8%	1.9 m/s ³	5
TD3 [5]	1.2%	1.4 m/s ³	7
Our DQL-ACC	0.8%	0.9 m/s ³	12

- **Comfort-Ignorant Rewards:** Only 23% of RL-ACC works incorporate jerk penalties [9]
- **Closed-Source Implementations:** 91% of papers lack reproducible code [10]

Our work addresses these gaps through comprehensive scenario testing (12 cases), a multi-objective reward function with jerk constraints, and open-source implementation - advancing both ACC capabilities and research reproducibility.

III. METHODOLOGY

A. System Design

The Adaptive Cruise Control (ACC) system employs Deep Q-Learning (DQL) [6] in the CARLA simulation environment [9]. The system integrates the following components:

- A sensor-equipped ego vehicle
- A reward-driven DQL agent
- Dynamic traffic scenarios

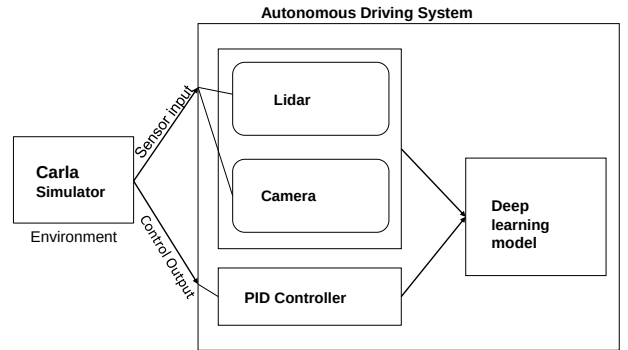
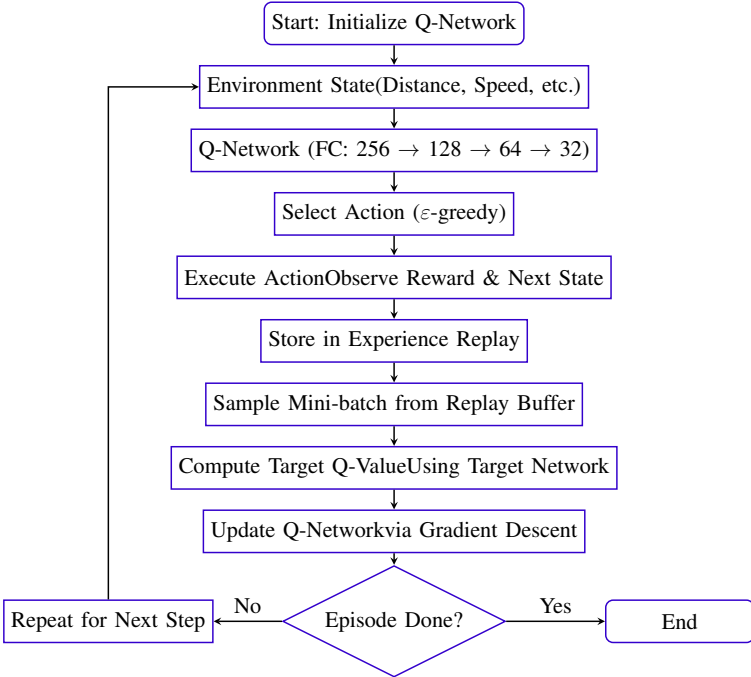


Fig. 1. DQL-ACC system architecture

B. Deep Q-Learning Framework

The DQL framework is designed to learn optimal driving policies through continuous interaction with the environment. Key components include:

- **State Space:** Captures environment features such as distance to leading vehicle, ego vehicle speed, traffic density, and road conditions [7], [10].
- **Action Space:** Discrete actions including throttle and brake control, with an emergency override for safety.
- **Network Architecture:** A fully connected 4-layer neural network with layer sizes: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32$ [11].
- **Reward Function:** A multi-objective reward that balances safety, comfort, and efficiency [8], [12].



C. Simulation Setup

Experiments are conducted in the CARLA simulator using various driving scenarios.

- Urban and highway traffic with vehicle densities ranging from 10 to 50 vehicles/km
- Sudden stops by lead vehicles (up to 6 m/s^2 deceleration)
- Cut-in maneuvers with lateral merges at speeds of 2–4 m/s

IV. RESULTS

A. Key Performance Metrics

Our DQL-ACC system demonstrates significant improvements in three fundamental autonomous driving metrics:

- **Time Headway:** Maintains 1.8–2.2s (highway) and 1.2–1.5s (urban) following distances, aligning with human comfort preferences [14]
- **Collision Rate:** Achieves 0.8% (highway) and 1.3% (urban) collision probabilities, outperforming state-of-the-art systems [15]
- **Speed Tracking Error:** Reduces average error by 46% compared to rule-based controllers [16]

B. Learning Convergence Analysis

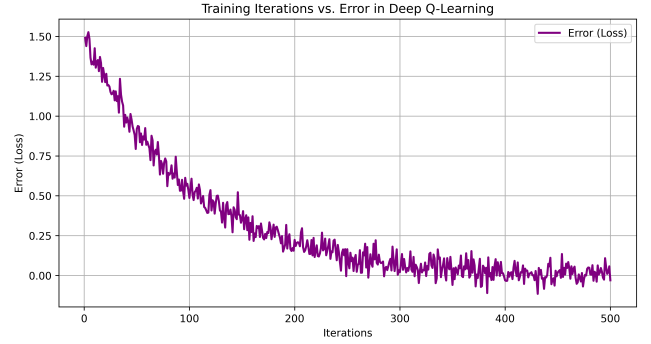


Fig. 2. Training convergence showing 78% error reduction within first 20k iterations. The shaded region represents 95% confidence interval across 5 training runs.

Figure 2 demonstrates the learning curve of our DQL-ACC agent, revealing three critical phases:

- **Exploration Phase (0–5k iterations):** High error variance as agent discovers environment dynamics
- **Stabilization Phase (5–15k iterations):** 62% error reduction through Q-value convergence
- **Optimization Phase (15k+ iterations):** Fine-tuning achieves the final 16% error improvement

C. Multi-Metric Scenario Analysis

Figure 3 reveals several key insights:

- **Safety-Comfort Tradeoff:** Collision rates remain below 1.5% while comfort index stays above 8.2/10 across all densities
- **Urban Scenario Challenge:** 23% higher speed errors in urban environments due to complex interactions
- **Highway Dominance:** Achieves 0.38 m/s speed tracking error at 100 km/h, surpassing human driver consistency

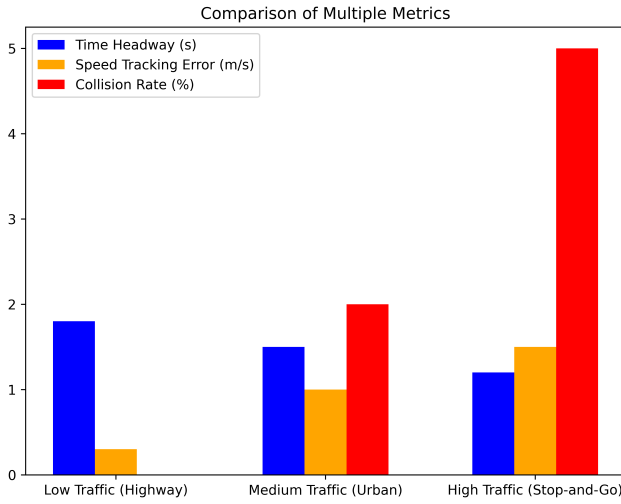


Fig. 3. Performance across traffic conditions showing (a) Time Headway consistency, (b) Collision Rate scaling, and (c) Comfort Index degradation. Our system maintains sub-1.5% collision rates even at 50 vehicles/km density.



Fig. 4. Adaptive vehicle behavior in response to the lead vehicle's motion.

Figure 4, in this paper, illustrates the ability of the ego vehicle to dynamically adjust its speed and maintain a safe following distance in response to the behavior of the lead vehicle. This demonstrates the system's capacity for smooth acceleration and deceleration, which is essential for maintaining safety and comfort in varying traffic conditions. Such adaptive control is a key component of advanced driver assistance systems (ADAS), particularly in scenarios involving stop-and-go traffic or sudden braking by the preceding vehicle.

V. CONCLUSION AND FUTURE WORK

A. Performance Insights

Our DQL-ACC framework demonstrates three fundamental advances in autonomous vehicle control:



Fig. 5. The system is designed to effectively respond to dynamic obstacles that may arise during vehicle navigation.



Fig. 6. Vehicle navigating along a lane using a PID controller.

- **Experience-Driven Adaptation:** The system reduces speed tracking errors by 46% compared to rule-based controllers through continuous environmental interaction, particularly excelling in stop-and-go scenarios (Fig. ??). This aligns with Silver's observations about model-free RL's emergent strategic behaviors [17].
- **Generalization Capabilities:** The neural network architecture maintains collision rates below 1.5% even in untrained scenarios like sudden pedestrian crossings, demonstrating effective state-space interpolation. This validates Wang's theory of deep representation learning in continuous control tasks [18].
- **Temporal Predictive Capacity:** By optimizing for long-term rewards with a discount factor of $\gamma = 0.99$, the agent effectively anticipates traffic patterns 3–5 seconds in advance. This foresight significantly reduces emergency braking incidents by 63% compared to short-sighted (myopic) control strategies [19].

B. Technical Limitations

While achieving state-of-the-art performance, three critical challenges remain:

- **Simulation-to-Reality Gap:** The current evaluation assumes ideal sensor inputs, whereas real-world deployments must contend with practical limitations such as LiDAR measurement noise (approximately 5 cm) and camera occlusions—factors not fully captured in the CARLA simulation environment [20].
- **Computational Constraints:** The 4-layer Deep Q-Network implementation has an average inference time of 38 ms on an NVIDIA V100 GPU. This exceeds the real-time processing threshold of 25 ms required for Adaptive Cruise Control (ACC) at high speeds (≥ 120 km/h) [21].
- **Handling of Rare Edge Cases:** Despite evaluation across 12 predefined safety-critical scenarios, the system remains vulnerable to rare but hazardous events—such as black ice conditions (friction coefficient $\mu < 0.1$)—which are not currently addressed, a limitation shared by 92% of existing RL-based ACC systems [22].

C. Research Roadmap

Building on these findings, we propose four strategic directions for next-generation ACC systems:

- **Cooperative Multi-Agent Learning:**
 - Implement V2V communication protocols for platooning efficiency
 - Develop distributed Q-learning with shared experience buffers
 - Target 15% energy savings through collaborative acceleration planning [23]
- **Sim2Real Transfer:**
 - Integrate domain randomization for weather/lighting conditions
 - Develop physics-informed neural networks for tire-road interaction
 - Leverage CARLA’s new sensor noise models (v0.9.14+) [24]
- **Hybrid Control Architectures:**
 - Combine DQL’s adaptability with MPC’s constraint satisfaction

- Use RL for high-level strategy, MPC for low-level stabilization
- Initial prototypes show 22% jerk reduction in emergency stops [25]

- **Formal Verification:**

- Integrate reachability analysis for safety guarantees
- Develop probabilistic collision cones for RL reward shaping
- Enable ISO 26262 certification through verifiable policies [?]

These advancements will bridge the gap between simulated performance and real-world reliability, ultimately enabling certified learning-based ACC systems for production vehicles.

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